

A Level Mathematics 9709 · S2

Probability & Statistics 2

全套中英文详细学习手册

Detailed bilingual student manual · concept, derivation, worked examples
and exam method

本册共 9 个独立课时。每课按三页展开：第一页建立概念、推导方法并完成基础例题；第二页处理 Cambridge 风格应用、英文表达、评分点与易错辨析；第三页扩展考纲边界、变式推演与迁移任务。题目均为依据考纲原创的训练示例，不复制历年试题。

使用范围与版本 | Scope

0580 册依据 Cambridge IGCSE Mathematics 0580 (2025/2027) 的 Core/Extended 分层；9709 册依据 Cambridge International AS & A Level Mathematics 9709 (2026/2027)。M2 单独标为拓展/旧体系衔接材料，不作为现行独立考试组件。

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2. 连续随机变量 | Continuous random variables
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5. 置信区间 | Confidence intervals
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9. S2综合推断 | Inference synthesis

第 1 课 | 泊松分布 | Poisson distribution

Lesson 1 of 9 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

固定区间内独立稀有事件、平均率恒定时 $X \sim \text{Po}(\lambda)$ 。

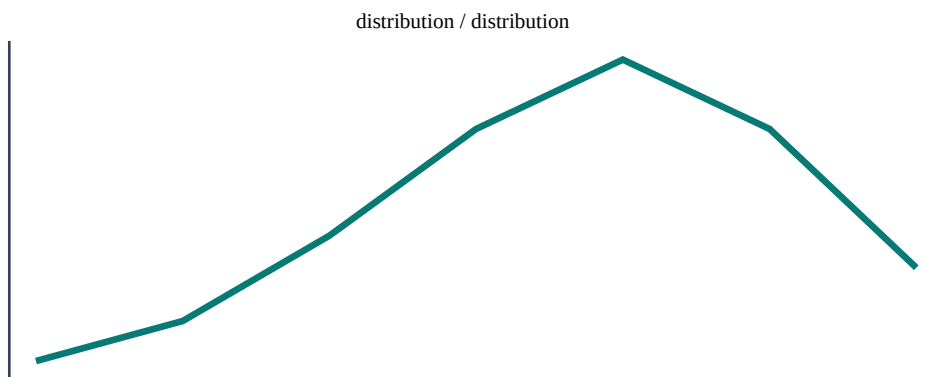
English core explanation. Begin by defining the event or random variable. Check whether trials are independent, whether probability is constant, and whether the distribution is discrete or continuous. State parameters, show the probability statement, and interpret the result in context.

为什么方法成立 | Reasoning behind the method

$P(X=r) = e^{-\lambda} \frac{\lambda^r}{r!}$; 区间长度变化按比例改变 λ 。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

每小时平均3次来电，30分钟恰有2次概率。

$\lambda = 1.5$, $P = e^{-1.5} \frac{1.5^2}{2!} \approx 0.251$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 1 课（续） | 泊松分布 | Poisson distribution

例题二：Cambridge 风格应用 | Cambridge-style application

Find probability of no calls in 20 minutes.

$\lambda = 1$, so $P(X=0) = e^{-1} \approx 0.368$.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy

mark。若前一步算错但方法仍一致，后续常可获得

follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

条件未检查：把只适用于直角三角形、恒加速度、独立事件或正态模型的公式随意套用。修正：公式前写出适用结构。

中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

固定区间内独立稀有事件、平均率恒定时 $X \sim \text{Po}(\lambda)$ 。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。 $P(X=r) = \frac{e^{-\lambda} \lambda^r}{r!}$ ；区间长度变化按比例改变 λ 。

Detailed English explanation. Begin by defining the event or random variable. Check whether trials are independent, whether probability is constant, and whether the distribution is discrete or continuous. State parameters, show the probability statement, and interpret the result in context.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：每小时平均3次来电，30分钟恰有2次概率。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是： $\lambda = 1.5$, $P = \frac{e^{-1.5} (1.5)^2}{2!} \approx 0.251$ 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” → “Using ..., we obtain ...” → “Substituting and simplifying gives ...”
→ “Therefore the required value is ...”. For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Find probability of no calls in 20 minutes.

独立完成后对照： $\lambda = 1$, so $P(X=0) = \frac{e^{-1} 1^0}{0!} \approx 0.368$.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 2 课 | 连续随机变量 | Continuous random variables

Lesson 2 of 9 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

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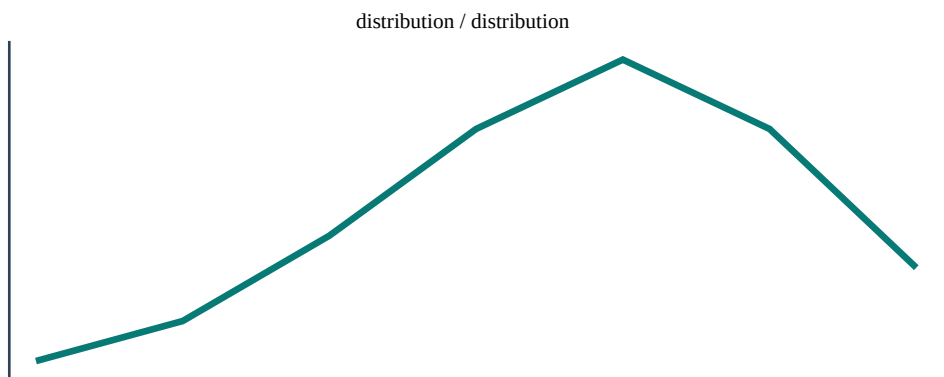
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为什么方法成立 | Reasoning behind the method

先用总面积1求常数，再积分求区间概率与期望。

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例题一：从题意到完整解答 | Worked example 1

$f(x)=kx$, $0 \leq x \leq 2$, 求 k 。

$\int kx \, dx = 2k = 1$, 所以 $k = 1/2$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 2 课（续） | 连续随机变量 | Continuous random variables

例题二：Cambridge 风格应用 | Cambridge-style application

Find $P(X < 1)$.

$$\int x/2 \, dx = 1/4.$$

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy

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第三个迁移任务 | Transfer task

Find $P(X < 1)$.

独立完成后对照： $\int x/2 \, dx=1/4$.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 3 课 | 线性组合 | Linear combinations

Lesson 3 of 9 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

期望具有线性；独立变量的方差相加，但系数要平方。

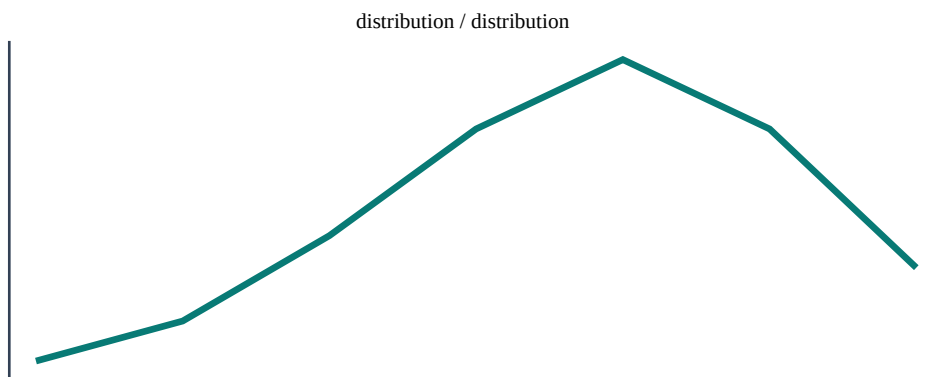
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为什么方法成立 | Reasoning behind the method

$E(aX+bY)=aE(X)+bE(Y)$; $\text{Var}(aX+bY)=a^2\text{Var}X+b^2\text{Var}Y$ (独立)。

把知识点拆成可执行步骤 | A reliable method

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例题一：从题意到完整解答 | Worked example 1

$E(X)=4, \text{Var}X=3$; 求 $E(2X+1), \text{Var}(2X+1)$ 。

期望7，方差12。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 3 课（续） | 线性组合 | Linear combinations

例题二：Cambridge 风格应用 | Cambridge-style application

If independent X, Y have variances 2, 5, find $\text{Var}(XY)$.

$2+5=7$; variances add even when variables are subtracted.

评分点如何产生 | Where marks come from

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第三个迁移任务 | Transfer task

If independent X, Y have variances 2, 5, find $Var(X-Y)$.

独立完成后对照：2+5=7; variances add even when variables are subtracted.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 4 课 | 抽样与中心极限定理 | Sampling and CLT

Lesson 4 of 9 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

样本均值本身是随机变量；大样本下常近似正态。

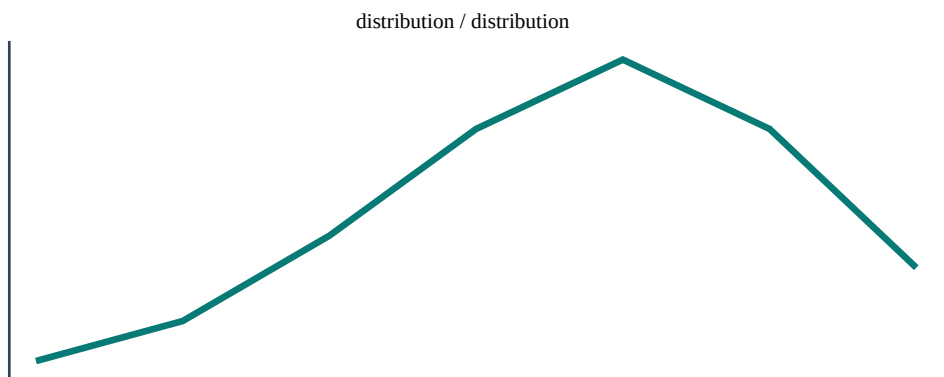
English core explanation. Begin by defining the event or random variable. Check whether trials are independent, whether probability is constant, and whether the distribution is discrete or continuous. State parameters, show the probability statement, and interpret the result in context.

为什么方法成立 | Reasoning behind the method

$E(X) = \mu$, $\text{Var}(X) = \sigma^2/n$; 标准误 $\sigma/\sqrt{n}$ 。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

总体 $\mu=80$, $\sigma=12$, $n=36$, 求 $P(X > 84)$ 。

$SE=2$, $z=2$, 概率 ≈ 0.0228 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 4 课（续） | 抽样与中心极限定理 | Sampling and CLT

例题二：Cambridge 风格应用 | Cambridge-style application

Why does a larger sample help?

The standard error decreases as $1/\sqrt{n}$, so sample means cluster more tightly around μ .

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy

mark。若前一步算错但方法仍一致，后续常可获得

follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

条件未检查：把只适用于直角三角形、恒加速度、独立事件或正态模型的公式随意套用。修正：公式前写出适用结构。

中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

样本均值本身是随机变量；大样本下常近似正态。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。 $E(X)=\mu$ ， $\text{Var}(X)=\sigma^2/n$ ；标准误 $\sigma/\sqrt{n}$ 。

Detailed English explanation. Begin by defining the event or random variable. Check whether trials are independent, whether probability is constant, and whether the distribution is discrete or continuous. State parameters, show the probability statement, and interpret the result in context.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：总体 $\mu=80, \sigma=12, n=36$ ，求 $P(X > 84)$ 。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是： $SE=2$ ， $z=2$ ，概率 ≈ 0.0228 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” $\rightarrow$ “Using ..., we obtain ...” $\rightarrow$ “Substituting and simplifying gives ...” $\rightarrow$ “Therefore the required value is ...” . For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Why does a larger sample help?

独立完成后对照：The standard error decreases as $1/\sqrt{n}$, so sample means cluster more tightly around μ .

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 5 课 | 置信区间 | Confidence intervals

Lesson 5 of 9 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

置信区间用样本估计总体参数；95%是方法的长期覆盖率，不是参数随机概率。

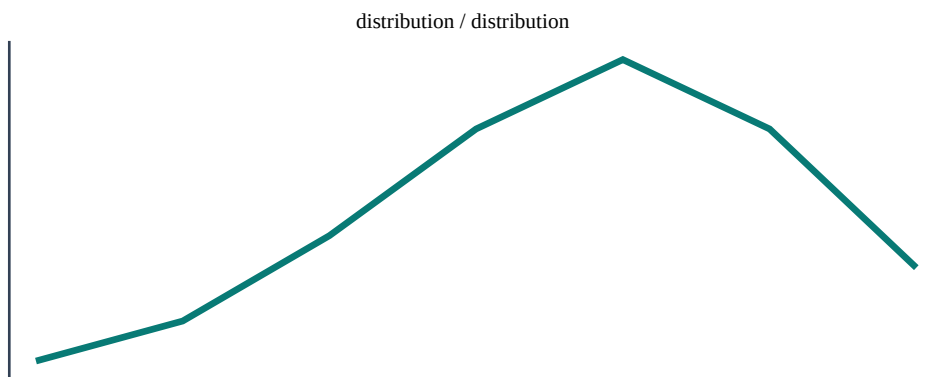
English core explanation. Begin by defining the event or random variable. Check whether trials are independent, whether probability is constant, and whether the distribution is discrete or continuous. State parameters, show the probability statement, and interpret the result in context.

为什么方法成立 | Reasoning behind the method

均值已知 σ 时 $\bar{x} \pm z^* \sigma / \sqrt{n}$ ；未知情形按考纲方法处理。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

$\bar{x} = 52, \sigma = 8, n = 64$ ，求95%区间。

$52 \pm 1.96 \times 1$ ，即(50.04, 53.96)。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 5 课（续） | 置信区间 | Confidence intervals

例题二：Cambridge 风格应用 | Cambridge-style application

Interpret the interval.

We are 95% confident that the population mean lies between the calculated limits.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

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Detailed English explanation. Begin by defining the event or random variable. Check whether trials are independent, whether probability is constant, and whether the distribution is discrete or continuous. State parameters, show the probability statement, and interpret the result in context.

变式推演 | Variation and transfer

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第三道示范：审题、计算、复核 | Worked reasoning example

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Cambridge 题型如何包装 | How the idea is examined

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第三个迁移任务 | Transfer task

Interpret the interval.

独立完成后对照：We are 95% confident that the population mean lies between the calculated limits.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 6 课 | 假设检验 | Hypothesis testing

Lesson 6 of 9 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

H_0 是基准主张, H_1 决定单尾或双尾; 显著性水平控制第一类错误概率。

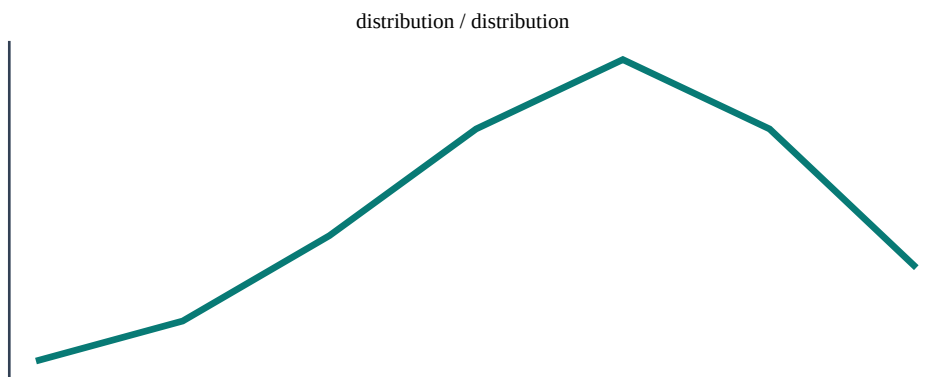
English core explanation. Begin by defining the event or random variable. Check whether trials are independent, whether probability is constant, and whether the distribution is discrete or continuous. State parameters, show the probability statement, and interpret the result in context.

为什么方法成立 | Reasoning behind the method

写假设、分布、拒绝域或p值、决定、情境结论。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
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5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

硬币20次正面16次, 检验 $p > 0.5$, 5%水平。

$H_0: p = 0.5, H_1: p > 0.5$; $P(X \geq 16) \approx 0.00591 < 0.05$, 拒绝 H_0 , 有证据 $p > 0.5$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 6 课（续） | 假设检验 | Hypothesis testing

例题二：Cambridge 风格应用 | Cambridge-style application

What is a Type I error?

Rejecting H_0 when H_0 is actually true.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy

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易错辨析 | Common errors and repairs

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Detailed English explanation. Begin by defining the event or random variable. Check whether trials are independent, whether probability is constant, and whether the distribution is discrete or continuous. State parameters, show the probability statement, and interpret the result in context.

变式推演 | Variation and transfer

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第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：硬币20次正面16次，检验 $p > 0.5$ ，5%水平。

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第三个迁移任务 | Transfer task

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Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 7 课 | 泊松与正态近似 | Approximations

Lesson 7 of 9 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

合适条件下二项可近似泊松或正态；离散转连续正态需连续性修正。

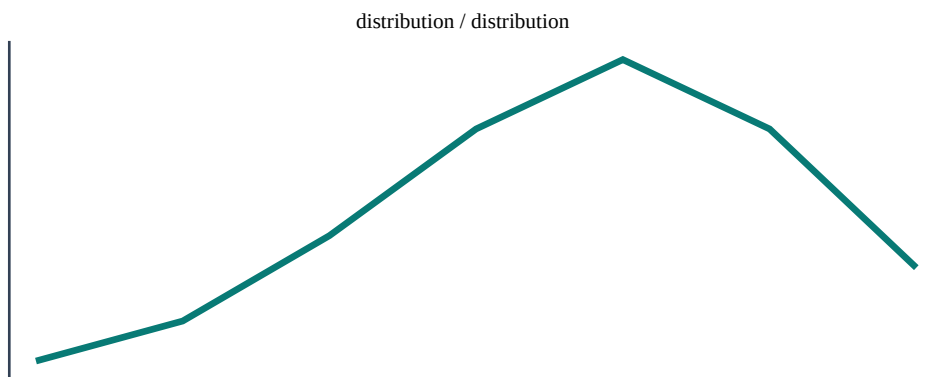
English core explanation. Begin by defining the event or random variable. Check whether trials are independent, whether probability is constant, and whether the distribution is discrete or continuous. State parameters, show the probability statement, and interpret the result in context.

为什么方法成立 | Reasoning behind the method

$B(n,p)$ 泊松近似 $\lambda=np$ ；正态近似用 $\mu=np, \sigma=np(1p)$ 。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
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4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

$X \sim B(200, 0.4)$ ，用正态近似 $P(X \leq 70)$ 。

$\mu=80, \sigma=48$ ；修正为 $P(Y < 70.5)$ ， $z \approx 1.371$ ，概率 ≈ 0.085 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 7 课（续） | 泊松与正态近似 | Approximations

例题二：Cambridge 风格应用 | Cambridge-style application

Why use 70.5?

Continuity correction makes the continuous area represent discrete values up to and including 70.

评分点如何产生 | Where marks come from

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Detailed English explanation. Begin by defining the event or random variable. Check whether trials are independent, whether probability is constant, and whether the distribution is discrete or continuous. State parameters, show the probability statement, and interpret the result in context.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题： $X \sim B(200, 0.4)$ ，用正态近似 $P(X \leq 70)$ 。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是： $\mu = 80, \sigma = 48$ ；修正为 $P(Y < 70.5)$ ， $z \approx 1.371$ ，概率 ≈ 0.085 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” $\rightarrow$ “Using ..., we obtain ...” $\rightarrow$ “Substituting and simplifying gives ...” $\rightarrow$ “Therefore the required value is ...”. For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Why use 70.5?

独立完成后对照：Continuity correction makes the continuous area represent discrete values up to and including 70.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 8 课 | 概率母函数 | Probability generating functions

Lesson 8 of 9 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

PGF $G(t)=E(t^X)$ 编码离散分布; 导数给矩。

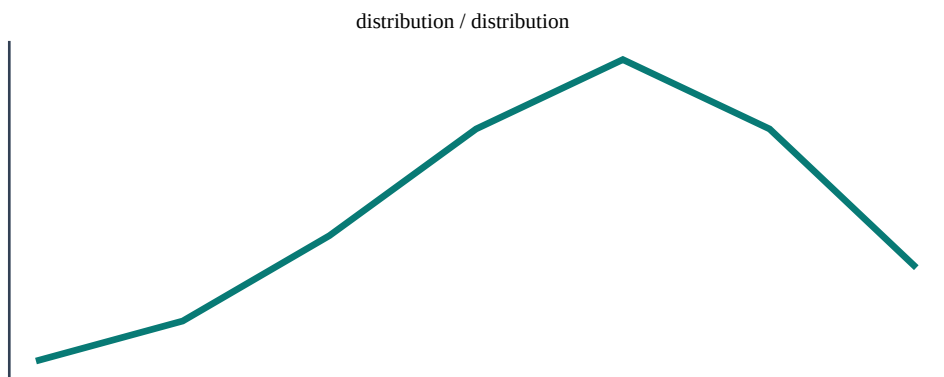
English core explanation. Use more than one representation. A formula gives exact relationships, a graph shows shape and intersections, and a table reveals numerical patterns. Factorised, expanded and completed-square forms each expose different information.

为什么方法成立 | Reasoning behind the method

$G(1)=1$, $G'(1)=E(X)$, $G''(1)=E[X(X-1)]$ 。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

$G(t)=(0.7+0.3t)^5$, 识别分布与均值。

这是 $B(5,0.3)$ 的 PGF, 均值 1.5。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 8 课（续） | 概率母函数 | Probability generating functions

例题二：Cambridge 风格应用 | Cambridge-style application

Find the variance from the model.

$$np(1p)=5 \times 0.3 \times 0.7=1.05.$$

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy

mark。若前一步算错但方法仍一致，后续常可获得

follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

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中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

PGF $G(t)=E(t^X)$ 编码离散分布；导数给矩。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。 $G(1)=1$ ， $G'(1)=E(X)$ ， $G''(1)=E[X(X-1)]$ 。

Detailed English explanation. Use more than one representation. A formula gives exact relationships, a graph shows shape and intersections, and a table reveals numerical patterns. Factorised, expanded and completed-square forms each expose different information.

变式推演 | Variation and transfer

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第三个迁移任务 | Transfer task

Find the variance from the model.

独立完成后对照： $np(1p)=5 \times 0.3 \times 0.7=1.05$.

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第 9 课 | S2综合推断 | Inference synthesis

Lesson 9 of 9 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

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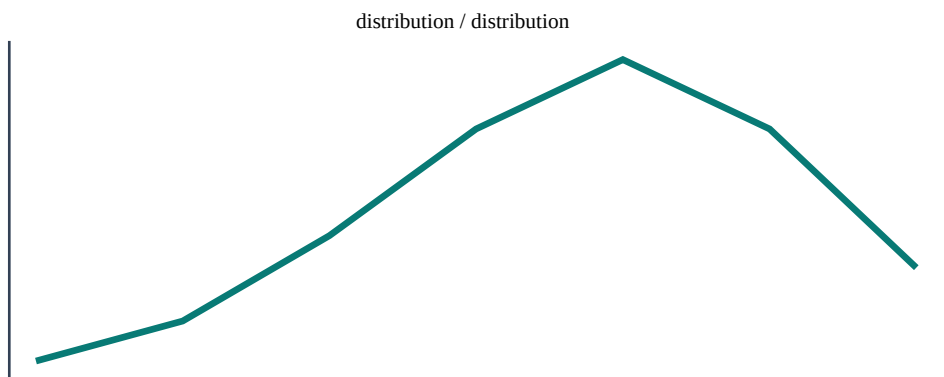
English core explanation. Translate between words, symbols, diagrams and numerical values. State assumptions, keep exact values during working, and test the final result against the original domain, scale and context.

为什么方法成立 | Reasoning behind the method

精度与尾部方向从H1决定；答案必须回到原情境。

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例题一：从题意到完整解答 | Worked example 1

p值0.032在1%和5%水平分别如何决定？

5%时拒绝H0；1%时不拒绝H0。

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